.

B. Tech CSE AIML & IoT (Ill YEAR - VI SEM) (2025-2026)

DEPARTMENT OF COMPUTER ENGINEERING & APPLICATIONS

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<u>Project Title</u>: AI-Powered Product Condition Verification for Safer E-Commerce

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Mentor Name: Mr. Shivanshu Upadhyay

Signature:

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Project Synopsis: <u>TrustLens AI</u> – AI-Powered Product Condition Verification for Safer E-Commerce

0. Cover

- **Project title:** AI-Powered Product Condition Verification for safer E-Commerce
- **Team name & ID:** Team VisionGaurd
- **Institute / Course:** GLA University, Mini-Project
- **Version:** v1.0
- **Date:** 04 Feb 2026

1. Overview

- **Problem statement:** Current e-commerce platforms lack transparent condition verification for damaged or refurbished products. Buyers cannot judge usability from vague "damaged" labels, and sellers struggle to price items fairly, leading to low trust and high dispute rates.
- **Goal:** Build an AI-powered system that analyses product images to detect damage, estimate severity, generate human-readable explanations, and recommend fair prices—bridging the trust gap in ecommerce.
- **Non-goals:** Image fraud detection, seller trust scoring, market-based dynamic pricing, and AI-assisted dispute resolution in MVP.
- **Value proposition:** Automated damage detection via CNN-based damage detection (YOLOv8 or equivalent lightweight model), severity scoring (1-10), clear condition explanations, and rule-based recommendations—all delivered within 5 seconds via a React web interface.

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2. Scope and Control

2.1 In-scope

- Image upload (multiple product images)
- AI-based damage detection using CNN-based damage detection (YOLOv8 or equivalent lightweight model)
- Severity scoring (1–10) with rule-based logic
- Simple condition explanation (Rule-based + template-driven explanation generation (LLM optional for future scope))
- Rule-based price recommendation
- Basic seller dashboard (upload → view result)
- Buyer result page (damage map, severity, explanation, price)
- Email-verified login/signup (optional if time permits)
- Image storage (local/S3)
- REST APIs for all core features
- Basic admin/moderation for test images
- Logging + error handling

2.2 Out-of-scope

- Market-based dynamic pricing (scraped or API-based)
- Advanced fraud detection (EXIF/ELA/reverse search)
- Seller trust scoring
- Full chatbot for interactive buyer queries
- Advanced dispute resolution
- Mobile apps (Android/iOS)
- High scalability / load balancing
- Payment systems or order workflows

2.3 Assumptions

- Product category is limited to **electronics** (mobile phones, laptops, accessories).
- All users are within project testing group (college team / mentor).

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- Dataset will be collected manually + online sources (Kaggle, Google images).
- Single backend server is used (Flask).
- No real-time GPU server; inference may run on CPU or Collab API.

2.4 Constraints

- 12–13 week mini-project timeline.
- Limited dataset (500–1000 images).
- Limited GPU access (Collab free tier).
- Beginner/intermediate skill level in ML + full stack.
- Deployment on free-tier services only (Vercel, Render, Supabase).
- YOLOv8 inference cannot exceed heavy server load.

2.5 Dependencies

- **SMTP service** (optional) for email-based verification.
- **Object storage** for images: AWS S3 / Supabase Storage / Firebase.
- **External APIs** (optional): Rule-based + template-driven explanation generation (LLM optional for future scope).
- for explanation module.
- **Model Training**: Google Collab for YOLOv8 training.
- **Environment**: Python 3.10+, React, SQLite (SQL).

2.6 Acceptance criteria and sign-off

- **AI Damage Detection:**
  GIVEN a user uploads a product image
  WHEN YOLO runs
  THEN bounding boxes + damage types must be returned in **≤ 5 seconds**.
- **Severity Score:**
  GIVEN YOLO outputs
  WHEN severity module runs
  THEN a severity value (1–10) is generated with correct mapping.
- **Price Recommendation:**

- GIVEN severity score and base price
  WHEN pricing engine runs
  THEN recommended price must match defined rule-based logic.
- **Explanation Module:**
  GIVEN severity + detected damage
  WHEN LLM summary is generated
  THEN user receives a **2–3 sentence** clear explanation.
- **Frontend Result Page:**
  GIVEN backend response
  THEN UI displays damage map, severity, explanation, and price correctly.

Sign-off table

Stakeholder	Role	Decision area	Signature/Approval	Date
Mr. Shivanshu Upadhyay	Mentor	Scope, final acceptance	Pending	TBD
Vivek Sharma	Project Lead	Release readiness	Pending	TBD

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3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Vivek Sharma	Vivek Sharma	Mr. Shivanshu	team
AI/ML Development	Shivansh Singh	Shivansh Singh	Mr. Shivanshu	Team
Deployment	Vivek Sharma	Vivek Sharma	—	Team
FrontEnd	Vivek Sharma	Vivek Sharma	Mr. Shivanshu	Team
BackEnd	Shivansh Singh	Shivansh Singh	Mr. Shivanshu	Dept

4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Vivek Sharma	Project Lead	FrontEnd & Deployment	React, Tailwind CSS, JavaScript, AWS	8 hrs/wk	vivek.sharma_cs.aiml23@gla.ac.in

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Shivansh Singh	UI/UX Developer	Backend & AIML Developer	Python, Flask, ML, NLP, Node.js	10hrs/wk	shivansh.singh_cs.aiml23@gla.ac.in
Shivansh Singh	AIML Developer	Damage Detection model, Severity Score model	ML, DL, open CV, NLP, Python	10 hrs/wk	shivansh.singh_cs.aiml23@gla.ac.in

5. Week-wise Plan and Assignments

Week	Dates	Milestone	Vivek (Lead)	Shivansh (Backend)	Vivek (Frontend)	Deliverables	Status
1	5-11 Feb	Project setup & planning	Setup dev environment, Flask skeleton	Dataset research & collection	React project init,wireframes	Project plan, dev setup	Planned

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2	12-18 Feb	Database Preparation	API endpoints design, image preprocessing module	Label Dataset augmentation pipeline	UI mockups ,component structure	Labeled Dataset, API spec	Planned
3	19-25 Feb	Model training begins	Flask upload endpoint, file handling	YOLOv8 training on damage detection	Image upload component, file validation	Initial model v0.1	Planned
4	26 – 3 Mar	Model integration	Integrate YOLOv8, inference pipeline	Model finetuning, hyperparameter optimization	Results display component	Working backend API	Planned
5	4-10 Mar	Severity & pricing logic	Severity estimation algorithm	Condition explanation module	Damage visualization overlay	Severity scoring module	Planned
6	11-17 Mar	Pricing engine	Rule-based pricing engine	Testing model accuracy, edge cases	Price recommendation UI	Pricing module complete	Planned

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7	18-24 Mar	Frontendbackend	API refinement,	Model perform ance	Full UI integratio n, state	End-toend flow working	Planned
		integratio n	error handling	optimiza tion	manage ment		
8	25–31 Mar	Testing & bug fixes	Unit tests, API documenta tion	Model validatio n on test set	UI/UX improve ments, responsiv eness	Test results, bug list	Planned
9	2-8 Apr	Deploym ent preparati on	Docker	Final model evaluati on report	Deploym ent setup and configura tion	Dockeriz ed applicati on	Planned
10	9-15Apr	User testing	Performanc e monitoring, logging	User feedbac k analysis	UI fixes based on feedback	User test report	Planned
11	16-22 Apr	Documen tation & polish	API docs, README	Technic al docume ntation	User guide, demo video	Complet e documen tation	Planned
12	23-28 Apr	Final presentat ion prep	Presentatio n slides	Demo preparat ion	Final testing, polish	Presentat ion, demo	Planned

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6. Users and UX

6.1 Personas

- **New Seller Sam:** Wants to upload product images quickly and get an instant AI condition report + fair price.
- **Buyer Bella:** Wants clear, trustworthy condition explanations and honest pricing before deciding to buy.

6.2 Top user journeys

- **Seller Journey**
  Home → Upload Image → Enter Base Price → Get AI Analysis Damage Map + Severity + Explanation + Recommended Price shown immediately.
  KPI: Process completes in ≤ 4 steps, success rate ≥ 85%.

- **Buyer Journey**
  Home → Open Product → View AI Condition Report → View Recommended Price → Decide to buy.
  KPI: Report loads in ≤ 2 sec, clarity rating ≥ 80%.

6.3 User stories Seller

- **Story:** As a seller, I want to upload product images and get an AI condition report in under 2 minutes so I can price and publish my listing quickly.
- **Acceptance:** GIVEN I am logged in, WHEN I upload images and submit, THEN I see the AI damage map, severity, explanation, and recommended price. **Buyer**
- **Story:** As a buyer, I want a clear AI-generated condition summary so I can decide whether the product is worth buying.
- **Acceptance:** GIVEN I open a product page, WHEN the AI report loads, THEN I see damage, severity, and price recommendation within 2 seconds.

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6.4 Accessibility & localization

- Basic keyboard navigation on main flows (upload, view report).
- UI follows contrast AA standards.
- Language: English only for v1.
- RTL or regional languages **not required** for mini-project.

7. Market and Competitors

7.1 Competitor table

Competitor	Product	Target users	Key features	Pricing	Strengths	Weaknesses	Our differentiator
OLX	C2C market place	General public	Listings, chat	Free	Large base	Ad clutter	Campus-verified users
Cashify	Electronics resale	India users	Multi-category	Free	Trusted brand	Manual condition checks only	Fully automated AI condition verification

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Refurb vendors (Amazon Renewed, Flipkart Renewed)	Refurb electro nics	Online buyers	Refurb checks	Free	High trust	Not available for individuals	Open to anyone; AIpowered instead of manual checks

7.2 Positioning

Unique Angle

AI-driven, transparent, and automated product condition verification for used electronics.

Measurable Differentiator

- AI report generated in ≤ 10 seconds vs. manual checks taking hours/days.
- Automated severity scoring and fraud checks not available in other platforms.
- Clear explanation + fair pricing = higher buyer trust.

8. Objectives and Success Metrics

1. Processing Speed
   AI damage detection + severity scoring + pricing should complete in **≤ 5 seconds** by final release.
   KPI: Median processing time (ms).

2. Explanation Quality

- At least **80% of users** (testers/mentors/classmates) should rate the AI-generated condition explanation as **“clear and helpful.” KPI:** User clarity score.

3. Accuracy Goals
   YOLO damage detection accuracy **≥ 80%** and severity mapping accuracy **≥ 75%** on validation dataset.
   KPI: Model accuracy (%).

4. Reliability & UI Performance
   Result page should load within **≤ 2 seconds** after AI response.
   KPI: p95 page-load time.

9. Key Features

Feature	Description	Priority	Dependencies	Acceptance criteria
Image Upload	Upload 1–5 product images	Must	Storage	GIVEN images WHEN uploaded THEN processed without errors
AI Damage Detection	YOLOv8 detects scratches, cracks, dents	Must	YOLO model	GIVEN image WHEN analyzed THEN bounding boxes return ≤ 5 s
Severity Scoring	Converts detected damage into severity (1–10)	Must	Detection output	GIVEN detections WHEN scored THEN severity correct by logic
Price Recommendation	Rule-based price after severity discount	Must	Severity score	GIVEN base price WHEN processed THEN recommended price shown

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Condition Explanation	Simple AIgenerated description for buyers	Should	Rule-based + template-driven explanation generation (LLM optional for future scope).	GIVEN severity + damage WHEN explained THEN 2–3 sentence output
Results Dashboard	Shows damage map + severity + explanation + price	Must	Backend APIs	GIVEN API response WHEN opened THEN results visible ≤ 2 s
Profiles (Seller)	Basic seller info + past listings	Could	Auth	GIVEN profile URL WHEN opened THEN details visible
Moderation (Basic)	Hide/report invalid uploads	Could	Auth	GIVEN report WHEN submitted THEN flagged in system

10. Architecture

10.1 High-level

- **Clients:** React SPA (image upload, result page, dashboard)
- **Services:**
- Auth service
- AI Analysis service (damage detection, severity scoring, price engine)
- Explanation service (Rule-based + template-driven explanation generation (LLM optional for future scope))
- Results service (stores + retrieves analysis reports)

- **Data Stores:**

- SQL (product info, results, users)
- Object storage (images) – AWS S3 / Supabase Storage
- **Integrations:**

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    - LLM API (OpenAI/Claude)
    - Optional SMTP for verification
    - Optional GPU/Colab for YOLO model inference

10.2 API spec snapshot

Endpoint	Method	Auth	Purpose	Request schema	Response schema	Codes
/api/auth/register	POST	—	Create account	email, password	201 {id}	201, 400
/api/upload	POST	JWT	Upload images	images[]	200 {fileUrls[]}	201, 400
/api/analyz e	POST	JWT	Run YOLO + severity + pricing	imageUrls[], basePrice	200 {damage, severity, price, explanation}	200,400, 500
/api/report/{id}	GET	JWT	Get past analysis	id	200 {fullReport}	200, 404
/api/listings	POST	JWT	Create listing with AI results	title, images[], aiReport	201 {listingId}	201, 400, 401
/api/search	GET	__	Browse listings	q, category, page	200 {items[], total}	200

10.3 Config and secrets

- **Environment variables:** Stored in. env, never committed to Git
- **API Keys:**
- OpenAI/Claude key
- Database credentials
- Storage bucket key

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- **Rotation:** Rotate LLM + SMTP keys each term
- **CI/CD:** Restrict CI pipeline access to maintain secret safety

11. Data Design

11.1 Data dictionary

Entity	Field	Type	Null ?	Allowed Values	Source	Notes
User	id	UUID	No	—	System	PK
User	email	String	No	RFC 5322	User	Unique
User	password_hash	String	No	—	System	Encrypted
Analysis Report	id	UUID	No	—	System	PK
Analysis Report	image_urls	JSON[]	No	URLs	System	Stored in S3
Analysis Report	damage_ data	JSON	No	YOLO output	AI Model	Boxes + labels
Analysis Report	severity	Integer	No	1–10	AI Model	Scored value

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Analysis Report	**explanation**	**Text**	**No**	**—**	**LLM**	**Summary text**
Analysis Report	**recommended_price**	**Decimal(10,2)**	**No**	**≥ 0**	**Pricing Engine**	**Final price**
Listing	**id**	**UUID**	**No**	**—**	**System**	**PK**
Listing	**title**	**String(120)**	**No**	**—**	**Seller**	**Indexed**
Listing	**base_price**	**Decimal(10,2)**	**No**	**≥ 0**	**Seller**	**Input**
Listing	**report_id**	**UUID**	**No**	**—**	**System**	**FK → Analysis Report**

11.2 Schemas and migrations

- ER diagram included in Appendix.
- Database migrations versioned.
- Rollback tested on staging environment.

11.3 Privacy, retention, backup/DR

- **PII:** name, email.
- **Retention:** delete accounts & reports after 12 months of inactivity.
- **Backups:** SQLite (SQL).

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12. Technical Workflow Diagram

i. State Transition Diagram

– Shows transitions from *Image Upload → Analysis Running → Results Ready → Listing Created*

ii. Sequence Diagram

– Step-by-step interaction between User → Frontend → Flask → YOLO Model → Severity Engine → Pricing Module → LLM → Database → Frontend

iii. Use Case Diagram

– Actors: *Seller, Buyer, AI System*
– Use cases: *Upload Image, Run Analysis, View Report, Create Listing, Browse Listing*

iv. Data Flow Diagram (DFD)

– Flow: *Images → Preprocessing → YOLO Detection → Severity → Pricing → Explanation → Database → UI* v. ER Diagram
– Entities: *User, Listing, Analysis Report*
– Relationships shown using PK–FK mapping vi. Technical Workflow Diagram

– End-to-end pipeline for AI processing and backend handling vii. Work

Architecture Diagram

– High-level architecture covering Frontend, Backend, AI Models, Storage, Database, APIs

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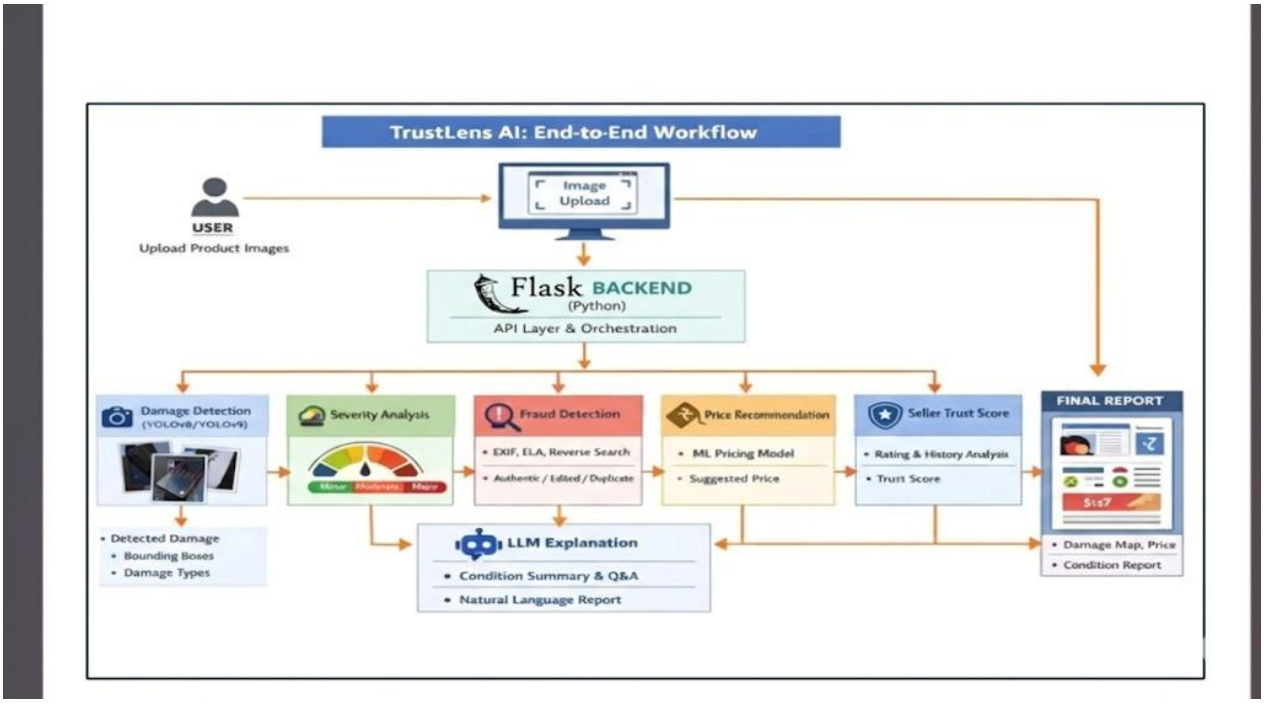
TrustLens AI: End-to-End Workflow
USER
Upload Product Images
Image Upload
Flask BACKEND
(Python)
API Layer & Orchestration
Damage Detection
(YOLOv8/YOLOv9)
• Detected Damage
• Bounding Boxes
• Damage Types
Severity Analysis
Fraud Detection
• EXIF, ELA, Reverse Search
• Authentic / Edited / Duplicate
Price Recommendation
• ML Pricing Model
• Suggested Price
Seller Trust Score
• Rating & History Analysis
• Trust Score
FINAL REPORT
• Damage Map, Price
• Condition Report
LLM Explanation
• Condition Summary & Q&A
• Natural Language Report

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13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	SLI	Target (SLO)	Measurement
Processing Speed	YOLO + severity + price time	≤ 5000 ms	APM / logs
UI Load Time	p95 load time	≤ 2000 ms	Browser profiling
Error Rate	5xx %	≤ 1%	Backend logs
Security	Open critical CVEs	0	Dependency scanner
Model Reliability	Damage detection accuracy	≥ 80%	Validation dataset

13.2 Test plan

Area	Type	Tools	Owner	Coverage Target	Exit Criteria
Backend	Unit Tests	PyTest	Team	70%	No P1/P2 defects
UI	E2E Tests	Playwright	Team	60%	Pass rate ≥ 95%
API	Integration Tests	Postman/Newman	Team	All critical scenarios	All critical pass
AI Model	Validation Tests	CNN-based damage detection (YOLOv8 or equivalent lightweight model)	Team	≥ 80% mAP	Accuracy target met

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13.3 Environments

- **Dev → Staging → Prod** pipeline • Feature flags for risky model updates
- Staging must pass: ❖ All API tests
  ❖ All E2E flows
  ❖ Basic model accuracy check

14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
Auth tokens	Token theft	Spoofing	High	Medium	HTTPS, short lived JWT, token rotation	Team
Uploaded images	Malware / Fake image injection	Tampering	Medium	Low	File-type validation, virus scan, size limits	Team
AI Model API	Model misuse / prompt injection	Elevation of Privilege	Medium	Low	Input sanitization, rate limiting	Team
User data	SQL injection	Tampering	High	Low	Parameterized queries, WAF	Team

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14.2 Audit and logging

- Log:
  ✓Signups
  ✓ Logins
  ✓ Image uploads
  ✓ AI analysis requests
  ✓ Listing create/update/delete
- Retention: **90 days**
- Sensitive logs: masked (email, IP)

14.3 Compliance

- Academic project; follow institute policies.
- No third-party sharing of user data.
- AI outputs tested for fairness + transparency.
- Uploaded images stored securely (private bucket).

15. Delivery and Operations

15.1 Release plan

- **v1.0 Mini-Project Demo:** Scheduled for **TBD** • Initial release includes: ✓ Image upload
  ✓ Damage detection
  ✓ Severity scoring
  ✓ Pricing engine
  ✓ Explanation module
- AI improvements (fraud detection, trust score) behind **feature flags**
- Optional: Dark-launch YOLO model improvements

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15.2 CI/CD and rollback

CI Pipeline:

lint → unit tests → build → dockerize

- **CD Pipeline:**

  Deploy to staging on each merge; manual promotion to production

- **Rollback Strategy:**

  Revert to previous working Docker image tag

15.3 Monitoring and alerting

Metric	Threshold	Alert to	Runbook
p95 AI processing latency	> 6000 ms	Tech Lead	“AI Latency” runbook
Error rate (5xx)	> 2%	Tech Lead	“Error Spike” runbook
Model failure rate	> 5%	ML Lead	“Model Failure” runbook

15.4 Runbooks

AI Latency Runbook

- Check YOLO model server load
- Verify storage (S3) response times
- Scale backend instance if needed
- Revert recent deployment if regression suspected

Error Spike Runbook

- Inspect backend logs
- Identify failing endpoint
- Roll back to previous version
- Document issue in incident log

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Model Failure Runbook

- Validate model input format
- Check version mismatch
- Switch to fallback model
  Escalate to ML lead

16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Score	Mitigation	Owner	Status
Model accuracy lower than expected	Medium	High	12	Data augmentation, hyperparamet er tuning	Team (ML Lead)	Open
Slow AI processin g time	Medium	Medium	9	Optimize YOLO model, use batching, reduce image size	Backen d Lead	Open
High storage usage for images	Low	Medium	6	Compression, size limits, auto-delete old images	Team	Open
LLM cost overruns	Low	Medium	6	Caching responses, smaller models for MVP	Team	O

17. Research and Evaluation

- **Market Research:**
  Studied OLX, Cashify, and FB Marketplace to understand trust gaps and how damaged products are handled.
- **Technical Research:**
  Benchmarked YOLOv8 damage detection models, LLM summarization quality, and rule-based pricing logic.
- **Evaluation Plan:**
- Track KPIs weekly (processing time, accuracy, clarity score).
- After MVP release, run a user survey on 10–15 testers.
- **Limitations:**
  No payment gateway, no delivery system, single-campus test environment, basic moderation only.

18. Appendices

- **Glossary:**
  YOLO, mAP, LLM, JWT, KPI, SLO, p95, CVE, inference time.
- **References:**
  Ultralytics docs (YOLOv8), FastAPI docs, React docs, PostgreSQL docs, academic ML papers on defect detection.